MOHAMMED IRSHAD C

Python Developer | Django, Backend & Full-Stack Systems

Malappuram, Kerala, India • +91 98461 35537 • irshaduu01@gmail.com

github.com/Irshaduu • linkedin.com/in/mohammed-irshad-c

SUMMARY

Python developer who ships production systems end to end. Sole engineer on a 49-model Django / PostgreSQL platform (169,000 lines across 510 files) built to replace paper job cards, WhatsApp coordination and manual spreadsheets at a premium car workshop. Nine months of work, 3,185 automated tests; tested on staging and approved for go-live by owners who had tried off-the-shelf workshop software since opening without finding a fit. Two further products built solo and deployed: a live multiplayer game and a rehabilitation app used daily through a real recovery. Ran a business before writing code, so costing, profit and cash flow are modelled as three separate figures.

TECHNICAL SKILLS

Languages	Python · JavaScript (ES6+) · SQL · HTML5 · CSS3
Frameworks	Django 5.2 · Django Channels (WebSockets, ASGI) · Django ORM · Bootstrap
Databases	PostgreSQL · SQLite · schema design · migrations · query optimisation (N+1 elimination, composite indexing)
Frontend	Vanilla JavaScript · PWA · Service Workers · Web Push (VAPID) · Web Audio API · Web Speech API · mobile-first
APIs & Integrations	REST · JSON APIs · WebSockets · AWS SigV4 request signing · Cloudflare R2 (S3-compatible)
Testing & Quality	Django test framework · 3,400+ automated tests (Django + JS) across projects · regression and invariant testing
Infrastructure	Railway · Render · environment-split settings · automated backup rotation · Git / GitHub
Engineering	System architecture · RBAC · decimal-based financial logic · audit trails · async & concurrency · auth hardening

EXPERIENCE

Software Developer (Freelance), Formula D, Pullara, Kerala

2026 – Present

WorkshopOS (Titan): Django / PostgreSQL platform (private client repository) — System map · Technical overview

- Engaged by a fast-growing premium car workshop running on physical job cards, WhatsApp coordination and manually maintained spreadsheets. Mapped how the business works on the floor, then built one system to replace all three: job cards, inventory, supplier ledgers, payments, payroll, estimates, warranty claims and three-tier role-based access.
- Delivered **49 models, 324 routes and 140 screens** in 294 commits (Jan – Oct 2026), backed by **3,185 automated Django tests** plus 71 JavaScript tests at a 1.34 : 1 test-to-code ratio.
- Built a **weighted-average inventory costing engine** as a date-ordered replay, chosen over FIFO so negative stock and back-dated supplier bills re-price correctly; per-batch cost is retained so FIFO stays reconstructible.
- Isolated **21 rule modules that hold no views**, each owning one question (profit and cash, invoicing, settlement, money parsing), so no figure can be derived two different ways. Money movements are protected by **72 all-or-nothing transaction blocks** and a Change History audit trail.
- Hardened access with per-account and per-IP lockout, device-aware session tracking with remote revocation, and OTP reset. Added Web Push over VAPID and direct-to-object-store photo uploads signed with stdlib HMAC against AWS SigV4, verified against Amazon's published test vector.

Technical Consultant (Freelance), Badusha Event Management

2026

Catering staff management system (build pending client budget) — Architecture report · Module plan

- Wrote the architecture and product specification for a 10-role staff-management PWA to replace paper and WhatsApp coordination: 24 modules tiered MVP / Optional / Deferred across a six-app Django structure, written so other developers can build from it.
- Flagged India's DPDP Act risk for Aadhaar data and designed the retention policy to show at upload (retention period pending legal check).

PROJECTS

Word Hunt: Real-Time Multiplayer Game Engine — word-hunt-game.onrender.com

- Multiplayer game on **Django Channels / ASGI** with a six-state server-authoritative state machine. A dropped WebSocket marks a player disconnected rather than removing them, so a mobile network blip never costs the time bank the game is scored on; reconnection uses exponential backoff with full state resync.
- Runs at **$0/month**: an in-memory channel layer removes the Redis dependency, and every in-game sound is synthesised with the Web Audio API, removing audio assets entirely. Server-side rate limiting and allowlisting on broadcast events; 78 tests.

Therapist: Offline-First Bilingual Medical PWA — therapist.pythonanywhere.com

- Hands-free post-operative knee rehabilitation app, built for a real recovery and **used daily through rehabilitation**. A localStorage sync queue replays sessions after network loss and the Screen WakeLock API keeps the display awake mid-exercise, so the patient never touches the phone during a routine. 80 tests.
- Engineered a **three-tier audio fallback** (Web Speech API, then 215 pre-generated English/Malayalam clips, then an on-demand server endpoint), including a stall timeout that clears the browser's known hung-utterance bug: reliable voice cues at zero recurring API cost.

EDUCATION & CERTIFICATIONS

Python / Django Full-Stack Development | Entri Elevate · Grade A+ (98%) · 2025

Python Programming | Certified User, Illinois Institute of Technology (08/2025) · Grade A+, Entri Software / Skill India NSDC (04/2025)

EARLIER BACKGROUND

Founder, Event Management Company (2019–2023). Built and scaled an events business, later handed to family. Four years of P&L, staffing and supplier ownership, applied directly to the systems now being built.

Algorithmic Trading (2023–2024). Deployed and managed automated trading systems; hands-on risk management.